START

System Setup
- Serial & I2C Init
- Wi-Fi & Blynk Init
- MAX30100 Init
- DHT22 Init
- DS18B20 Init
- OLED Init
- LED & Buzzer Pins

Loop Starts

Update MAX30100
(Read HR & SpO_2)

MAX30100 Responsive?

NO

Reset I2C & Re-initialize MAX30100

Wi-Fi Connected?

YES

NO

Run Blynk

Offline Mode

Read Sensors
- DHT22 (Room Temp & Humidity)
- DS18B20 (Body Temp)

Averaging Countdown Started?

YES

NO

Valid HR & SpO_2 Detected? (Finger Placed)

YES

NO

Start Countdown & Averaging (20s)

Measurement Complete?

YES

NO

Compute Averages
- Avg Heart Rate
- Avg SpO_2

Send Averaged Data to Blynk
(If Wi-Fi Connected)

Alarm Condition?
HR < 60bpm or
HR > 100bpm
Body Temp > 37.2°C
Room Temp > 32°C

YES

NO

LED ON
Buzzer ON
Send Alerts

UPDATE OLED DISPLAY
- Room Temperature
- Room Humidity
- Body Temperature
- Show HR & SpO_2 (only if available)
- Show “Place Finger” if HR not valid
- Status Messages & Alerts

Periodic Blynk Update (Every 10 Seconds)
(If Wi-Fi Connected)

Loop Repeats